AP CALCULUS AB

Unit 1 • Worksheet 1-4A • Answer key

Worksheet 1-4A • Continuity at a Point

Answer key with worked solutions

Directions

Answer every question. Multiple-choice questions have one best answer. On free-response questions, show your work and justify your answers. Questions labeled **NO CALCULATOR** must be done without a calculator; questions labeled **CALCULATOR** require a graphing calculator. For questions with a graph, read the values you need from the grid.

Question 1

NO CALCULATOR

Let $f(x) = \begin{cases} x^2 + 1, & x < 1 \\ 3x - 1, & x \geq 1. \end{cases}$ Which is true?

(A) $\lim_{x \to 1} f(x)$ does not exist.

(B) The limit exists but f is not continuous at $x = 1$.

(C) f is continuous at $x = 1$.

(D) $f(1)$ is undefined.

Solution

Answer: (C). Limits 2 and 2, $f(1) = 2$.

Question 2

NO CALCULATOR

For what value of k is $f(x) = \begin{cases} kx + 3, & x \leq 2 \\ x^2 - k, & x > 2 \end{cases}$ continuous?

(A) $-\frac{1}{3}$

(B) $\frac{1}{3}$

(C) 1

(D) 7

Solution

Answer: (B). $2k + 3 = 4 - k$.

Question 3

NO CALCULATOR

What type of discontinuity does $g(x) = \dfrac{x^2 - 4}{x - 2}$ have at $x = 2$?

(A) jump

(B) infinite

(C) removable

(D) none

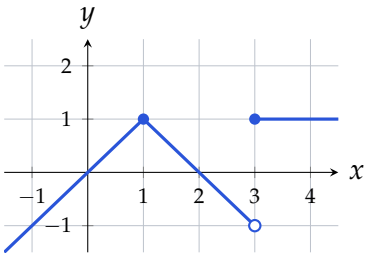
Solution

Answer: (C).

Question 4

NO CALCULATOR

The graph of f is shown. Which statement is true?



(A) continuous at $x = 3$ but not at $x = 1$
(B) continuous at $x = 1$ but not at $x = 3$
(C) continuous at both
(D) continuous at neither

Solution
Answer: (B).

Question 5

NO CALCULATOR

On which interval is $h(x) = \dfrac{\sqrt{x+2}}{x-1}$ continuous?

(A) $(-2, \infty)$
(B) $[-2, 1) \cup (1, \infty)$
(C) $(1, \infty)$
(D) all real numbers except 1

Solution
Answer: (B).

Question 6 (Free response)

NO CALCULATOR

Let $f(x) = \begin{cases} 2x + a, & x < 1 \\ bx^2 + 1, & x \geq 1 \end{cases}$ and $f(2) = 9$.

(a) Find b, then the value of a that makes f continuous at $x = 1$.
(b) Verify continuity at $x = 1$ using the definition.

Solution
(a) $4b + 1 = 9$, $b = 2$; $2 + a = 3$, $a = 1$. (b) $f(1) = 3$; both one-sided limits are 3; so $\lim_{x \to 1} f(x) = f(1)$.

Question 7 (Free response) NO CALCULATOR

Let $h(x) = \dfrac{x^2 + x - 6}{x^2 - 4}$. Find each discontinuity, classify it, and justify with limits. What value of $h(2)$ would remove the removable one?

Solution
$h = \frac{(x+3)(x-2)}{(x+2)(x-2)}$. At $x = 2$: $\lim h = \frac{5}{4}$, removable; define $h(2) = \frac{5}{4}$. At $x = -2$: $\lim_{x \to -2^-} h = -\infty$, $\lim_{x \to -2^+} h = +\infty$: infinite.